Project Design Phase-II
Technology Stack (Architecture & Stack)

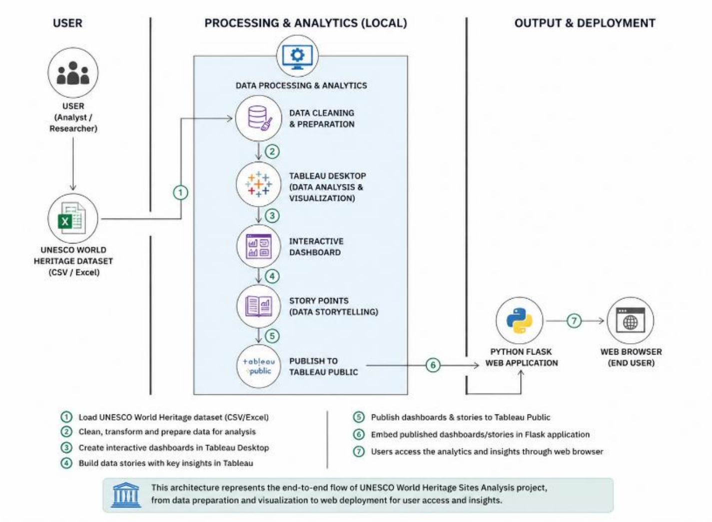
Date	01 July 2026
Team ID	
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- **Include all the processes (As an application logic / Technology Block)**
  Data Cleaning & Preparation, Data Analysis & Visualization (Tableau Desktop), Interactive Dashboards, Data Storytelling, Publish to Tableau Public, Deployment via Flask Web Application.
- **Provide infrastructural demarcation (Local / Cloud)**
  Local Processing & Analytics (Tableau Desktop) and Cloud Deployment (Tableau Public & Flask Web App).
- **Indicate external interfaces (third party API's etc.)**
  Web Browser (End User Access), Tableau Public (Cloud Platform).
- **Indicate Data Storage components / services**
  UNESCO World Heritage Dataset (CSV/Excel), Cleaned Dataset (Local Storage), Published Content (Tableau Public).
- **Indicate interface to machine learning models (if applicable)**
  Not applicable for this project (No ML models used).
- **Include data sources**
  UNESCO World Heritage Sites Dataset (CSV/Excel).
- **Include output & deployment components**
  Dashboards & Stories published to Tableau Public and embedded in Python Flask Web Application for end user access.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Provides an interactive web interface for users to explore UNESCO heritage dashboards and stories.	HTML5, CSS3, JavaScript.
2.	Application Logic-1	Loads UNESCO World Heritage dataset and performs preprocessing operations.	Python
3.	Application Logic-2	Performs data visualization and dashboard development.	Tableau Desktop
4.	Application Logic-3	Publishes dashboards and embeds Tableau Story into the web application.	Tableau Public, Python Flask
5.	Database	Stores UNESCO world Heritage dataset locally in CSV format for analysis.	Excel sheet.
6.	Cloud Database	Public cloud platform used to host published dashboards and stories.	Tableau Public Cloud.
7.	File Storage	Stores dataset, images, HTML templates, CSS, JavaScript files, and project resources.	Local Filesystem.
8.	External API-1	Embeds published Tableau dashboards into the Flask web application.	Tableau JavaScript API.
9.	External API-2	Serves static images and project assets through Flask.	Flask Static File Handling.
10.	Machine Learning Model	Forecasting future UNESCO Heritage trends using Tableau forecasting analytics.	Tableau Forecasting Model.
11.	Infrastructure (Server / Cloud)	Hosts the Flask web application and serves embedded Tableau dashboards through a local web server.	Python Flask (Localhost), Tableau Public.

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks and libraries used for web development and data processing.	Python Flask, Pandas, NumPy, HTML5, CSS3, JavaScript.
2.	Security Implementations	Protects application resources and ensures secure communication between Flask application and Tableau Public.	Flask Routing, HTTPS (Tableau Public), Input Validation.
3.	Scalable Architecture	Modular architecture separates data processing, visualization, and web interface, allowing future expansion.	Python Flask + Tableau Architecture.
4.	Availability	Dashboards are published online and accessible through Tableau Public, while the web application is available locally or can be deployed online.	Tableau Public, Flask Web Server.
5.	Performance	Optimized visualization and lightweight Flask application ensure fast loading and smooth interaction with dashboards.	Tableau Dashboard Optimization, Flask.

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d